

LINUX ASSIGNMENT -3

Q1. Difference between man and whatis commands with example. (CO1)

man command → It opens the manual (help book) of a command. It gives detailed usage, options, and explanation.

Example:

man ls

→ Shows full guide of ls.

whatis command → It shows only a short one-line description of the command.

Example:

whatis ls

→ Output: ls (1) - list directory contents.

So, man = full manual, whatis = short meaning.

Q2. Save output of ls -l into file and show it at the same time. (CO4)

ls -l | tee output.txt

This command shows the list of files on the screen and also saves the same result into output.txt.

Q3. Example of using tee command in logging. (CO4)

tee can be used to record outputs while still displaying them.

Example:

ping -c 3 google.com | tee log.txt

It prints the ping result on the screen and also writes it inside log.txt. This is useful

when we want to keep logs while monitoring live output.

Q4. Steps to install Ubuntu 25.04 LTS in VirtualBox. (CO2)

1. Download Ubuntu 25.04 LTS ISO file.
2. Open VirtualBox → click New and make a VM.
3. Choose type = Linux, version = Ubuntu (64-bit).
4. Assign memory (RAM) → around 2–4 GB.
5. Create a virtual hard disk (20 GB or more).
6. Go to Settings → Storage → add the Ubuntu ISO.
7. Start the VM → installer will begin.
8. Select language, keyboard, and installation type.
9. Enter username and password.
10. Finish setup and restart. Ubuntu will be ready.

Q5. How to solve Kernel Panic Error in Ubuntu installation? (CO3)

Check if ISO file is not broken (verify checksum).

Give enough RAM and CPU to the VM.

Enable virtualization (VT-x/AMD-V) in BIOS and VirtualBox.

Update VirtualBox version if needed.

If problem continues, try recovery mode or minimal installation.

Q6. Show hostname and change it using sysctl. (CO1)

To see current hostname:

```
hostname
```

To change hostname (temporary):

```
sudo sysctl kernel.hostname=newname
```

To make change permanent, edit the file `/etc/hostname`.

Q7. Command to show calendar of August 1984. (CO1)

```
cal 8 1984
```

Q8. Command to show system uptime and logged-in users. (CO3)

Option 1:

```
uptime && who
```

Option 2 (combined in one):

```
W
```

Q9. Command to find all `.c` files in `/home/user`. (CO1)

```
find /home/user -name "*.c"
```

Q10. Change file permission so only owner can read and write. (CO1)

```
chmod 600 filename
```

Here:

Owner → can read and write.

Group → no access.

Others → no access.

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